Def. Define

$$f_a^c : [-1,1] \to \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x=0 \\ x^a \cdot \sin(|x|^{-c}) & \text{if } x \neq 0 \end{cases}$$

where $a \in \mathbb{R}$, $c > 0$. And, define quotient at $x=0$ of f_a^c as

$$\varphi_a^c : [-1,1] \setminus \{0\} \to \mathbb{R} : t \mapsto \frac{f_a^c(t)}{t}$$

Then, $\varphi_a^c(t) = t^{a-1} \cdot \sin(|t|^{-c})$.

Prop.

$$\frac{d}{dx} f_a^c(x) = \begin{cases} \lim_{t \to 0} t^{a-1} \cdot \sin(|t|^{-c}) & \text{if } x=0, \text{ and the limit exists.} \\ a \cdot x^{a-1} \cdot \sin(|x|^{-c}) - c \cdot x^{a+1} \cdot |x|^{-c-2} \cdot \cos(|x|^{-c}) & \text{if } x \neq 0 \end{cases}$$

Simply, $a x^{a-1} \cdot \sin\left(\frac{1}{x^c}\right) - c \cdot x^{a-c-1} \cdot \cos\left(\frac{1}{x^c}\right)$ if $x > 0$

$2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$

$\lim_{t \to 0} t \cdot \sin$

$x^a \cos((x)^{-c}) \cdot -c \cdot x^{-c-1}$

$+ x^a \cdot \cos(|x|^{-c}) \cdot \frac{x}{|x|} \cdot (-c) \cdot |x|^{-c-1}$.

Equivalent Condition Table	f	f'	f''
Continuous	$a > 0$	$a > 1+c$	$a > 2+2c$
bounded	$a \geq 0$	$a \geq 1+c$	$a \geq 2+2c$
Exists at $x=0$	Always...	$a > 1$	$a > 2+c$
Bounded Variation			

「f_1

observe that $f: [-1,1] \to \mathbb{R}; x \mapsto \begin{cases} 0 & \text{if } x=0 \\ x\sin\left(\frac{1}{|x|}\right) & \text{if } x \neq 0. \end{cases}$

Then, f is Continuous, Not differentiable at $x=0$, Not bounded Variation, Not bounded derivative.

Proof. Since $\left|x\sin\left(\frac{1}{|x|}\right)\right| \leq |x| \to 0$ as $x \to 0$, Clearly continuous.

Meanwhile, $f'(0)$ exists $\iff \lim_{t\to 0} \sin\left(\frac{1}{|t|}\right)$ exists, thus not exists.

And, to show that f is Not bounded Variation, Construct a Partition as:

$$P_n \stackrel{\text{def}}{=} \left\{-1, 0, \frac{2}{n\pi}, \frac{2}{(n-1)\pi}, \cdots, \frac{2}{2\pi}, \frac{2}{\pi}, 1\right\}$$

Then, $V(f, P_{2n+1}) \geq \sum_{k=0}^{n-1} \frac{2}{(2k+1)\pi} = \frac{2}{\pi} \cdot \sum_{k=0}^{n-1} \frac{1}{2k+1} \to \infty$ as $n \to \infty$.

Therefore it is Not bounded variation. Similarly, f_n^n $(n \in \mathbb{N})$ are Not bounded Variation.

But, in this way, we can not determine $f_2'(x) = x^2\sin\left(\frac{1}{x}\right)$ is BV or Not.

observe that

$$\frac{d}{dx} f_2'(x) = \begin{cases} 0 & \text{if } x=0 \\ 2x\sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right) & \text{if } x>0 \end{cases}$$

It is Riemann integrable and bounded. Therefore, for any partition P on $[-1,1]$,

$$V(f_2', P) = \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| \underset{\text{MVT}}{=} \sum_{i=1}^{n} |f'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^{n} \sup|f'| \cdot \Delta x_i = \sup|f'| \cdot 2$$

Therefore, f_2' is Bounded Variation. Genellary,

「Prop. If $f: [a,b] \to \mathbb{R}$ is differentiable and the derivative f' is bounded, then f is bounded variation.

Proof. Ask a toddler on the street.

「Prop. f_2^2 is not bounded variation.

$$\frac{1}{x^2} = \left(\frac{1}{2}+n\right)\pi$$

$$x = \frac{1}{\sqrt{\left(\frac{1}{2}+n\right)\pi}}$$

Proof. Let $P_n = \{x_{-1}, x_0, \cdots, x_n, x_{n+1}\}$ as $x_{-1} = -1$, $x_{n+1} = 1$, and

$$x_k = \frac{1}{\sqrt{\left(\frac{1}{2}+n-k\right)\pi}} \qquad 0 \leq k \leq n$$

Then,

$$V(f_2^2, P_n) \geq \sum_{i=1}^{n} \left| f_2^2(x_i) - f_2^2(x_{i-1}) \right|$$

$$= \sum_{i=1}^{n} \frac{1}{\left(\frac{1}{2}+n-i\right)\pi} + \frac{1}{\left(\frac{1}{2}+n-(i-1)\right)\pi}$$

$$\geq 2 \cdot \sum_{i=1}^{n} \frac{1}{\left(\frac{1}{2}+n-i+1\right)\pi}$$

$$= \frac{2}{\pi} \cdot \sum_{i=1}^{n} \frac{1}{\frac{1}{2}+i} \geq \frac{2}{\pi} \sum_{i=1}^{n} \frac{1}{i} \to \infty \quad \text{as } n \to \infty$$